Symmetry of four-digit continued-fraction permutiples

Principia Math In Progress

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MathDB 335304; Holt, arXiv:1702.04760v1, Conjecture 17.

Theorem 1 (Task 1)**.** *Let a, b, c, d be positive integers, and let (e, f, g, h) be a permutation of (a, b, c, d). Suppose that $d, h \geq 2$ and*

$$[a; b, c, d] = k[e; f, g, h], \qquad k \in \mathbb{Z}, \quad k \geq 2.$$

Then $ad = eh$ and $bc = fg$. Thus every canonical four-digit continued-fraction permutiple is symmetric in Holt's sense.

Continuants and two reductions

Put

$$P = abcd + ab + ad + cd + 1, \qquad Q = bcd + b + d,$$

and define P', Q' using (e, f, g, h) in the same order. The continued fractions are P/Q and P'/Q', in lowest terms. Taking integer parts gives

$$a = ke + j, \qquad 0 \leq j \leq k - 1. \tag{1}$$

In particular $a > e$ and $k \leq a/e \leq a$.

Lemma 1. *For any two arrangements of four positive integers, their four-term continuants have ratio strictly less than two.*

Proof. Write the sorted entries as $1 \leq x \leq y \leq z \leq w$ and set $V = xyzw$. The continuant of an arrangement is $V + S + 1$, where S is the sum of the three edge products along a path through the four entries: for (a, b, c, d) the path is b, a, d, c. Root this path at x. Each other vertex v contributes an edge of weight at least xv, so $S \geq x(y + z + w)$. Also $S \leq w(y + z) + yz$. Indeed, if w has degree two its incident edges sum to at most $w(y + z)$ and the other edge is at most yz. If it has degree one its incident edge is at most wz and the other two sum to at most $yz + xz \leq yz + wy$. With $L_0 = V + x(y + z + w) + 1$ and $U_0 = V + w(y + z) + yz + 1$, direct expansion gives

$$\begin{aligned} 2L_0 - U_0 = (x - 1)\{yzw + 2(y + z + w)\} \\ + (y - 1)(z - 1)(w - 1) + y + z + w + 2 > 0. \end{aligned}$$

Every such continuant therefore lies between L_0 and $U_0 < 2L_0$. □

Since $PQ' = kP'Q$ and $\gcd(P', Q') = 1$, we have $P' \mid P$. Lemma 1 now gives

$$P = P', \qquad Q' = kQ. \tag{2}$$

For completeness, set $p_2 = abc + a + c$, $q_2 = bc + 1$ and define p_2', q_2' analogously. The determinant identities are

$$Pq_2 - Qp_2 = 1, \qquad Pq_2' - Q'p_2' = 1.$$

Subtracting them and using (2) yields

$$Q \mid (bc - fg), \qquad P \mid (p_2 - kp_2'). \tag{3}$$

Lemma 2. *If $h \geq k$, then $ad = eh$ and $bc = fg$.*

Proof. We have $0 < p_2 < P$ and $P = hp_2' + ef + 1 > hp_2'$, so $0 < kp_2' < P$ when $h \geq k$. By (3), their difference, which lies strictly between $-P$ and P, is zero. The subtracted determinant identities then imply $q_2 = q_2'$, or $bc = fg$. The equality $abcd = efgh$ gives $ad = eh$. □

Exhausting the permutations

The six positional permutations with $e = a$ are excluded by $a > e$. The six with $h = a$ satisfy Lemma 2, since $a \geq k$. The remaining twelve are in the following table. Set $L = b - c + d$ and $N = bc + bd - cd$. Every expression in the middle column must vanish by (2).

(e, f, g, h)	$P - P'$	Resolution
(b, a, c, d)	$d(a - b)$	$a = e$, impossible
(b, a, d, c)	$ad - bc$	symmetry directly
(b, c, a, d)	$b(a - c) - d(b - c)$	Case A
(b, d, a, c)	$aL - N$	Case B
(c, a, b, d)	$aL - bd$	Case C
(c, a, d, b)	$aL - N$	Case D
(c, b, a, d)	$b(a - c)$	$a = e$, impossible
(c, d, a, b)	$ad - bc$	symmetry directly
(d, a, b, c)	$b(a - c)$	$h = a$, Lemma 2
(d, a, c, b)	$b(a - c) - d(b - c)$	Case E
(d, b, a, c)	$aL - bd$	Case C
(d, c, a, b)	$d(a - b)$	$h = a$, Lemma 2

This is a list of positional permutations, so repeated digit values cause no exception to its exhaustiveness.

Case C. Here $aL = bd > 0$, so the integer L is at least one. Since $c = b + d - L$,

$$cL - b = (c - 1)(L - 1) + d - 1 \geq 0.$$

Thus $a = bd/L \leq cd = eh$. As $k \leq a/e$, it follows that $h \geq k$ and Lemma 2 applies.

A common bound for Cases B and D. If $aL = N$ and $L \geq 1$, then $a \leq bc$. For $b \leq c$, this follows from $N = bc - d(c - b) \leq bc$. For $b > c$, use $L \geq d + 1$ and $b - c < bc$ to obtain $N = bc + d(b - c) < (d + 1)bc \leq Lbc$. In both cases $eh = bc$, so $h \geq k$ and Lemma 2 applies.

Case D. Write $a = kc + j$ with $j \geq 0$. The equality $Q' = kQ$ is

$$kL = j(bd + 1) + b > 0.$$

Consequently $L \geq 1$ and the common bound proves symmetry.

Case B. The common bound handles $L \geq 1$. If $L = 0$, then $c = b+d$ and $N = b^2+bd-d^2 = 0$, which would give $(2d-b)^2 = 5b^2$, impossible for positive integers. Suppose $L < 0$ and put $u = -L \geq 1$, so $c > b + d > d$. Write $a = kb + j$ with $j \geq 0$. Using $Q' = acd + c + d = kQ$ and $aL = N$ gives

$$jQ = c(a-d), \qquad ad - bc = (a-d)(c-b), \qquad a - d = \frac{d^2 - bc}{u}. \tag{4}$$

The first identity implies $a \geq d$. If $a = d$, the second proves $ad = bc$ and hence symmetry. Otherwise $0 < a - d < d^2$. By (3), $Q \mid (ad - bc)$. The first two identities in (4) then imply $Q \mid b(a-d)$ as well as $Q \mid c(a-d)$. Writing $s = \gcd(b, c)$, Bézout's identity gives $Q \mid s(a-d)$. But

$$0 < s(a-d) < sd^2 \leq bd^2 < bcd < Q,$$

a contradiction. This completes Case B.

Case A. The numerator identity is $b(a-c) = d(b-c)$, so

$$a = c + d - \frac{cd}{b}, \qquad b(a-b) = (b-c)(d-b).$$

Since $a > b$, the possibility $d = b$ is excluded. If $d > b$, then $a < d = h$, and Lemma 2 applies. If $d < b$, the second identity forces $c > b$, and then $a < c$. Write $a = kb + j$ with $j \geq 0$. The denominator identity becomes

$$c(jd+1) = kb + (k-1)d.$$

If $j \geq 1$, then, because $c > a \geq kb$,

$$c(jd+1) \geq c(d+1) > kb(d+1) \geq k(b+d) > kb + (k-1)d,$$

which is impossible. If $j = 0$, then $a = kb$ and $c = kb + (k-1)d$. Substituting into $b(a-c) - d(b-c) = 0$ gives $(k-1)d^2 = 0$, again impossible.

Case E. Again $a = c + d - cd/b < c + d$. Write $a = kd + j$, $j \geq 0$. The denominator identity is

$$j(bc+1) = (k-1)b.$$

It forces $j \geq 1$ and hence $c < k - 1$. But then

$$a < c + d < k + d - 1 \leq kd \leq a,$$

a contradiction. All permutations have been resolved, proving the theorem.

Scope and reproducibility

The theorem answers Conjecture 17 for all positive digits and every integer multiplier $k > 1$, using the source's canonical conventions. No distinctness or bounded-digit assumption is made. It does not assert the longer-length conjectures. The proof uses only elementary continuant identities; no other structural theorem in the source is needed. The accompanying verifier checks the polynomial identities, the full permutation table, and bounded exact instances. The latter are checks, not a substitute for the proof. This is author-agent self-review, not independent peer review.

Proposed follow-up tasks

These are research directions, with no solutions claimed here.

2. Determine the sharp supremum of the ratio of two four-term continuants obtained by permuting one positive-integer multiset.

3. For each fixed integer $k > 1$, parametrize all canonical four-digit continued-fraction permutiples without repetitions.

4. Decide whether every canonical five-digit continued-fraction permutiple preserves its numerator continuant.

5. Classify the canonical five-digit permutiples whose two outer-pair products and middle digit are preserved by the permutation.

References

[1] B. V. Holt, *Digit Preserving Multiplication in Continued Fraction Representations*, arXiv:1702.04760v1 (2017), Definitions 1 and 13, Conjecture 17. `https://arxiv.org/abs/1702.04760`.